

Statistical Data Analysis

Lecture 6: The Bootstrap – part 2

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Periods 4 & 5

📍 Chapter 5: The bootstrap

- ◀◀ Section 5.1 – Simulation

- ◀◀ Section 5.2 – Bootstrap estimators for a distribution

- Section 5.3 – Bootstrap confidence intervals

- Section 5.4 – Bootstrap tests

- Section 5.5 – Limitations of the bootstrap

▶▶ Chapter 6: Nonparametric methods

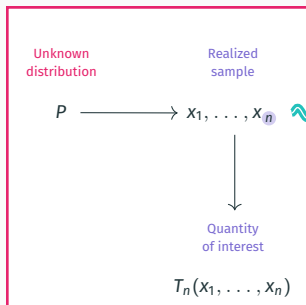
Bootstrap – Recap

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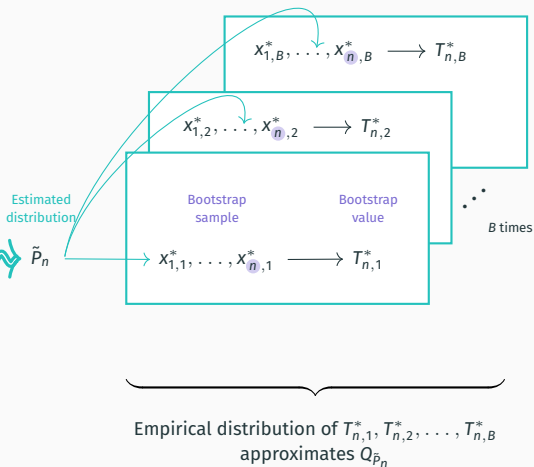
- › Data $X_1, \dots, X_n \stackrel{iid}{\sim} P$, where P **unknown**.
 - › Quantity of interest $T_n = T_n(X_1, \dots, X_n) \sim Q_P$.
 - › We are interested in properties of Q_P .
- › Bootstrap idea: Replace P by an estimator $\tilde{P}_n$ and Q_P with $Q_{\tilde{P}_n}$
 - › Given the sample $x_1, \dots, x_n$, estimate P by $\tilde{P}_n$.
 - › Simulate a bootstrap sample $X_1^*, \dots, X_n^*$ from $\tilde{P}_n$
 - › Compute a bootstrap value $T_n^* = T_n(X_1^*, \dots, X_n^*)$.
- › Types of bootstrap
 - › Empirical bootstrap: $\tilde{P}_n = \hat{P}_n$ is the empirical distribution (resampling)
 - › Parametric bootstrap: $\tilde{P}_n = P_{\hat{\theta}}$ is an estimated distribution

Bootstrap – Real World vs. Bootstrap World

Real World



Bootstrap World



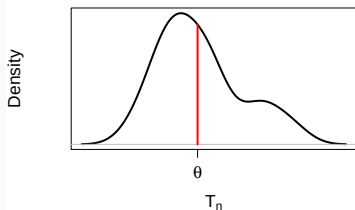
Bootstrap Confidence Intervals

Confidence Intervals – Recap

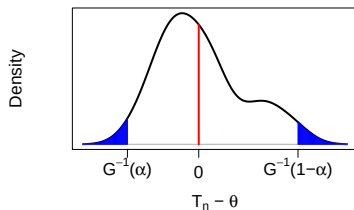
- › Accuracy of an estimate?
 - › Bias, variance (or standard deviation), confidence interval
- › Suppose T_n is an estimator for unknown θ
 - › Then, $(T_n - \theta) \sim G_n$
 - › By definition of quantiles: $\mathbb{P}(G_n^{-1}(\alpha) \leq T_n - \theta \leq G_n^{-1}(1 - \alpha)) \geq 1 - 2\alpha$
 - › Then, $\mathbb{P}(T_n - G_n^{-1}(1 - \alpha) \leq \theta \leq T_n - G_n^{-1}(\alpha)) \geq 1 - 2\alpha$
 - › $[T_n - G_n^{-1}(1 - \alpha), T_n - G_n^{-1}(\alpha)]$ is a $(1 - 2\alpha)$ confidence interval for θ
- › **Caveat:** for realized $T_n = t$ and corresponding realized confidence interval, it makes little sense to speak about probability of containing θ
 - › Once the interval bounds are computed (given t), the interval either contains θ or not
 - › For realized intervals, we rather talk about (long-run) relative frequency of coverage
 - › This relates to the reliability of the estimation procedure (a priori)
 - › θ is still unknown

Confidence Intervals – Illustration

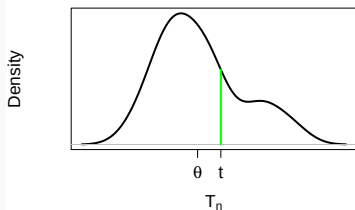
Q_P : distribution of T_n



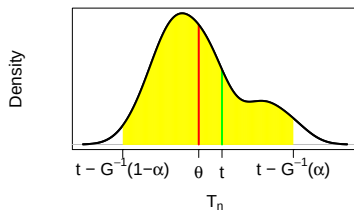
G : distribution of $T_n - \theta$



Q_P and realisation of T_n



realised conf.int. for θ



Bootstrap Confidence Intervals – Idea

- › In $[T_n - G_n^{-1}(1 - \alpha), T_n - G_n^{-1}(\alpha)]$ there are **unknowns**:
 - › G_n , i.e. the distribution of $T_n - \theta$
 - › Q_p , i.e. the distribution T_n
 - › θ , i.e. the parameter for which T_n is an estimator

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- › **Idea**: estimate G_n by the empirical distribution of **bootstrap** values
 - › These are $Z_i^* = T_{n,i}^* - T_n$, for $i = 1, \dots, B$
 - › $T_{n,i}^*$ obtained via **empirical** or **parametric** bootstrapping

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 - › These are $Z_i^* = T_{n,i}^* - T_n$, for $i = 1, \dots, B$
 - › $T_{n,i}^*$ obtained via **empirical** or **parametric** bootstrapping
- › Replace quantiles of G with **sample quantiles** of $Z_1^*, \dots, Z_B^*$
 - › Replace $G_n^{-1}(\alpha)$ with $Z_{([\alpha B])}^*$ and $G_n^{-1}(1 - \alpha)$ with $Z_{([(1-\alpha)B])}^*$
 - › In **R**: `quantile`
- › **Bootstrap confidence interval**:

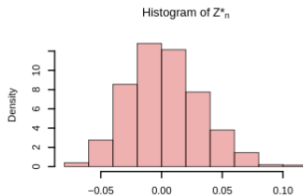
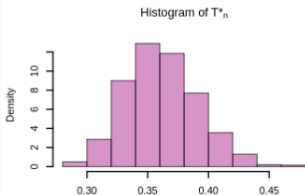
$$[T_n - Z_{([(1-\alpha)B])}^*, T_n - Z_{([\alpha B])}^*] = [2T_n - T_{n,([(1-\alpha)B])}^*, 2T_n - T_{n,([\alpha B])}^*]$$

- › The equality follows from the fact that $Z_i^* = T_{n,i}^* - T_n$ for $i = 1, \dots, B$

Bootstrap Confidence Intervals – Illustration

```
### Data: elevator waiting times from assignment 3
waits <- readRDS("waiting_times.RDS"); T_n <- 1/mean(waits) # Realized  $T_n$ 
```

```
### Bootstrap confidence interval for rate lambda?
set.seed(123); B <- 10^3; T_star <- numeric(B); Z_star <- numeric(B)
for(i in 1:B){
  X_star <- sample(waits, replace = TRUE)
  T_star[i] <- 1/mean(X_star)
  Z_star[i] <- T_star[i] - T_n
}
```



```
### Bootstrap confidence intervals
ci_T <- c(T_n - quantile(Z_star, 0.975), T_n - quantile(Z_star, 0.025))
sprintf("Bootstrap CI for lambda: [%.3f, %.3f]", ci_T[1], ci_T[2])

## [1] "Bootstrap CI for lambda: [0.295, 0.409]"

ci_Z <- c(2 * T_n - quantile(T_star, 0.975), 2 * T_n - quantile(T_star, 0.025))
sprintf("Bootstrap CI for lambda: [%.3f, %.3f]", ci_Z[1], ci_Z[2])

## [1] "Bootstrap CI for lambda: [0.295, 0.409]"

### AVOID (for this course): 'percentile bootstrap'
ci_perc <- quantile(T_star, c(0.025, 0.975))
sprintf("Percentile bootstrap CI for lambda: [%.3f, %.3f]", ci_perc[1], ci_perc[2])

## [1] "Percentile bootstrap CI for lambda: [0.309, 0.422]"
```

Bootstrap Tests

Bootstrap Tests – Idea

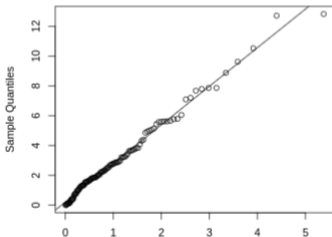
- › We have (realizations from) $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} P$
 - › P is **unknown**
- › **Aim:** goodness-of-fit test for testing $H_0 : P \in \mathcal{P}_0$ vs. $H_1 : P \notin \mathcal{P}_0$
 - › $\mathcal{P}_0$ is collection of distributions
- › Test statistic T_n has **unknown** null distribution $\rightsquigarrow$ p -value?
 - › Assuming T_n is **nonparametric** under $H_0 \rightsquigarrow$ **Bootstrap**
- › **Example:** $H_0 : P \in \{\text{Exp}(\lambda); \lambda > 0\}$
 - › Possible test statistic $T_n = \frac{\text{med}(X)}{\bar{X}} \sim Q_{P_0}$
 - › Q_{P_0} is **unknown** but independent of $\lambda \rightsquigarrow T_n$ ‘**nonparametric**’
 - › **Estimate** p -values via ‘**bootstrap**’

Bootstrap Tests – Illustration

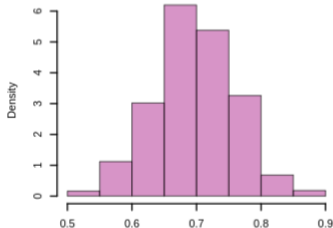
```
### Data: elevator waiting times from assignment 3
# But this time  $T_n = \text{median}/\text{mean}$  is a realized test statistic ( $T_n$  ca. = 0.73)
waits <- readRDS("waiting_times.RDS"); T_n <- median(waits)/mean(waits)
```

```
### Bootstrap
set.seed(123); B <- 10^3; T_star <- numeric(B); n <- length(waits)
for(i in 1:B){
  X_star <- rexp(n)
  T_star[i] <- median(X_star)/mean(X_star)
}
```

Exponential Q-Q Plot for Elevator Waiting Times



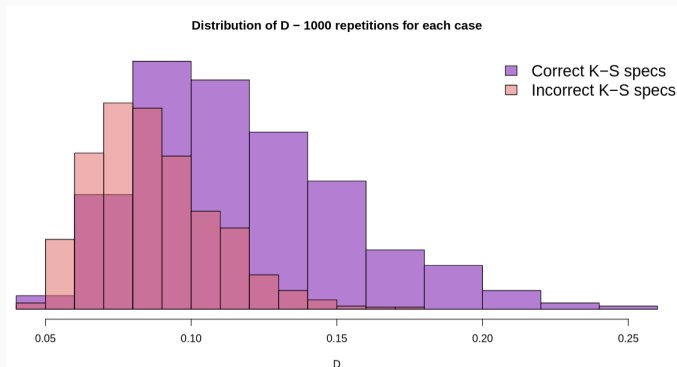
Histogram of T_n^*



```
### Bootstrap-based p-value (two-sided test)
p_boot <- 2 * min(mean(T_star <= T_n), mean(T_star >= T_n))
p_boot
```

```
## [1] 0.61
```

Bootstrap Tests – An Old Friend...



- Recall that we **reject** for **large** values of D ...
- $\tilde{D}_n$ (Syllabus) is a **sensible** test statistic for **composite** normality
- But p -value given by `ks.test(x, "pnorm", mean(x), sd(x))` is **wrong**
- It is calculated from the distribution of D_n (with **correct** specs)
- Estimate the distribution of $\tilde{D}_n$ via **bootstrap**
- 🔗 Nonparametric? See the syllabus and A4 😊

Limitations of the Bootstrap

Limitations of the Bootstrap

- **Warning:** the bootstrap can fail:
- Parametric bootstrap can fail if:
 - **Wrong** parametric family is chosen
 - There are important outliers in the sample
 - Parameter estimators are not accurate
- Empirical bootstrap can fail:
 - **Particularly** for extreme order statistics, e.g. $X_{(1)} = \min(X_1, \dots, X_n)$ or $X_{(n)} = \max(X_1, \dots, X_n)$
- Heavy-tailed data distribution are particularly **problematic** for bootstrap methods
 - See Example 5.6 in the Syllabus

Summary and wrap-up

› Bootstrap confidence intervals

- › Estimate θ by T_n
- › Estimate quantiles of the distribution G_n of $T_n - \theta$ from a bootstrap sample
- › Create $(1 - 2\alpha)\%$ -CI for θ

› Bootstrap tests

- › Estimate the nulldistribution of a **test statistic** T_n
- › Simulate bootstrap samples under the null hypothesis.

› Limitations of the Bootstrap

- › Parametric: wrong parametric assumptions, outliers, inaccurate estimators
- › Empirical: may fail for extreme order statistics
- › Heavy-tailed distributions are usually problematic.

What now?

› To do for now

- › Start with assignment 4
- › Deadline: 31 March 23:59
- › All material covered

› Next week

- › We are halfway through the course
- › Next week: discussion of a former midterm exam by Philip Boeken (available [here](#))
- › This is the level you should be at now!

› After that

- › No lecture / tutorial in the week of March 23rd – 27th
- › From april 1st: Time / location change of lectures, check rooster.vu.nl!